

TUNIR SIKDAR

Data Engineer II | Deloitte USI

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PROFESSIONAL SUMMARY

Data Engineer with 5 years of experience architecting and scaling distributed data platforms on AWS and Azure. Skilled at translating ambiguous data problems, missing feeds, inconsistent vendor formats, historical gaps, into fault-tolerant, idempotent pipeline designs and Lakehouse architectures supporting high-volume, mission-critical workloads. Owns system design end-to-end, improving reliability, performance, and cost efficiency for Fortune 500 clients across insurance, retail, and e-commerce.

SKILLS

- **Languages:** Python, SQL, T-SQL
- **System Design:** Distributed systems - Lambda & Kappa architecture, Medallion architecture, SLA-driven orchestration, scalability & process optimization
- **AWS:** Lambda, Glue, S3, EC2, CloudWatch, Secrets Manager
- **Azure:** Databricks, Data Factory (ADF), Data Lake Storage (ADLS Gen2), Key Vault, Azure SQL Database
- **Big Data:** Apache Spark, Apache Iceberg, Hadoop (basic)
- **Data Engineering:** Data modeling & normalization, Data Warehousing, Data Vault 2.0, data quality frameworks
- **DevOps & Tools:** Azure DevOps, Git, CI/CD

EXPERIENCE

Data Engineer II | Deloitte USI – Kolkata, India

04/2025 – Present

- Architected a greenfield, config-driven, idempotent ingestion framework to ingest vendor feeds into a single insurance data platform, including backfill logic to safely reprocess historical batch data for newly onboarded sources, scaling to 4.2 TB of daily batch volume with zero duplication or data loss on reprocessing.
- Implemented technical and functional data quality checks that routed malformed records to a dead-letter queue (DLQ) for isolation and reprocessing, and designed imputation and standardization rules to handle missing or malformed fields across third-party feeds.
- Led a team of 5 engineers to re-architect the commission payout calculation process, cutting processing time from a week to a daily cadence (reducing manual effort by ~90% through automated jobs), and implemented a commission chargeback workflow that reduced turnaround from one month to one day in computing the chargeback amounts owed by the business.
- Architected a fault-tolerant orchestration layer on AWS Lambda and SQL Server control tables with automatic retry and checkpointing, sustaining a 99%+ on-time SLA for critical batch processing, with SNS-based alerting for failures or warnings and a robust audit mechanism for full traceability.

Firm Initiatives

- Built an event ingestion and analytics pipeline for a mobile application using Azure Event Hubs and Spark Structured Streaming, powering dashboards that track each user's health criteria and capture longitudinal usage history, giving business stakeholders visibility into user trends over time.

Data Engineer | Fractal Analytics – Pune, India

09/2023 – 03/2025

- Delivered 24 end-to-end Big Data projects on Azure, designing scalable PySpark and Spark SQL jobs on datasets exceeding 500+ GB that improved execution efficiency by over 40%.
- Resolved recurring data gaps in a Fortune 500 retail client's inventory feeds by designing config-driven ETL pipelines with imputation logic for missing SKU-level fields and automated backfill jobs to reprocess historical records, structuring output into Delta tables with star and snowflake schema modeling for reliable BI reporting at scale.
- Reduced manual intervention in error handling by 60% by designing automated data validation and monitoring frameworks in PySpark, and cut annual licensing costs by \$150K by re-architecting workflows onto Azure.

Application Developer | IBM India Pvt. Ltd. – Kolkata, India

02/2021 – 08/2023

- Migrated 40+ business-critical pipelines from legacy systems to Azure using ADF and built data quality validation checks in Databricks and Python.
- Reduced manual data preparation time by 42 hours per month by streamlining extraction processes, and cut operating costs by 30% through faster issue resolution.

EDUCATION

B.Tech. in Electronics and Communication Engineering, Heritage Institute of Technology

08/2016 – 08/2020

GPA: 9.09/10.0

CERTIFICATIONS

- Databricks Certified Data Engineer Associate (2025)
- Fractal Certified PySpark Developer

AWARDS

- Outstanding Award & Applause Award, Deloitte USI | Star Award (Kaizen), Fractal Analytics | Manager Appreciation Award, IBM India Pvt. Ltd.